

A useful theorem for designing Controlled-U gates

$$CNOT = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X$$

$|0\rangle\langle 0|$

When no X in between:

$$\overbrace{R_z(\mu) R_y\left(\frac{\nu}{2}\right)}^A \overbrace{R_y\left(-\frac{\nu}{2}\right) R_z\left(\frac{-\xi-\mu}{2}\right)}^B \overbrace{R_z\left(\frac{\xi-\mu}{2}\right)}^C = I$$

$|1\rangle\langle 1|$

When X is in between:

$$\overbrace{R_z(\mu) R_y\left(\frac{\nu}{2}\right)}^A X \overbrace{R_y\left(-\frac{\nu}{2}\right) R_z\left(\frac{-\xi-\mu}{2}\right)}^B X \overbrace{R_z\left(\frac{\xi-\mu}{2}\right)}^C = R_z(\mu) R_y(\nu) R_z(\xi) = U$$

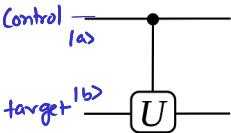
$$CU = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$$

$$CU|a\rangle|b\rangle = |0\rangle\langle 0|a\rangle \otimes |b\rangle + |1\rangle\langle 1|a\rangle \otimes U|b\rangle$$

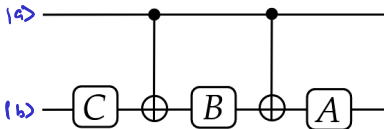
$$|\psi\rangle = \sum_{a=0}^1 \sum_{b=0}^1 \langle a,b | \psi \rangle |a\rangle|b\rangle$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & U \end{bmatrix}$$

$$(CU)^{-1} = (CU)^{\dagger} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U^{\dagger}$$



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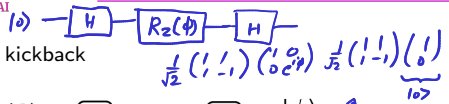
$\{H, T, CNOT\}$

$a, b \in \{0, 1\}$

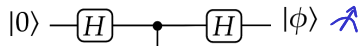
Quantum Interferometer

Important quantum subroutines

Phase kickback



Control



target eigenket



$U|u\rangle = e^{i\phi}|u\rangle$
↑ eigenvalue

$$|0\rangle|u\rangle \mapsto \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|u\rangle$$

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$$

$$X|\pm\rangle = \pm|\pm\rangle$$

$$\mapsto \frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle)|u\rangle$$

$$\frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle)|u\rangle$$

$$\mapsto \left(\frac{1+e^{i\phi}}{2}|0\rangle + \frac{1-e^{i\phi}}{2}|1\rangle \right)|u\rangle$$

$$e^{i\phi/2} \left[\cos(\phi/2)|0\rangle - i\sin(\phi/2)|1\rangle \right]|u\rangle$$

$$\{(\cos(\phi/2))^2, (\sin(\phi/2))^2\}$$

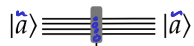
Uses: Quantum phase estimation, Shor algorithm, Quantum HHL algorithm for system of linear equations, Hadamard test

$$\cos(\phi/2) = \frac{e^{i\phi/2} + e^{-i\phi/2}}{2} \quad \sin(\phi/2) = \frac{e^{i\phi/2} - e^{-i\phi/2}}{2i}$$

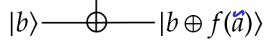
$$\begin{matrix} \text{---} X \text{---} \\ \oplus \end{matrix} = \begin{matrix} \text{---} X \text{---} \\ \oplus \end{matrix} = 1 \otimes X \otimes I + 1 \otimes I \otimes X$$

Boolean function evaluation in reversible (quantum) computation

Input Register



Output Register



$$a = a_1 a_2 \dots a_n \quad a_i \in \{0,1\}$$

$$|a\rangle = |a_1\rangle |a_2\rangle \dots |a_n\rangle \quad b \in \{0,1\}$$

a is such that $f(a)=0$
 $\tilde{a}$ is " " $f(\tilde{a})=1$

$$U_f|a\rangle|1\rangle = |a\rangle|1\rangle$$

$$U_f|\tilde{a}\rangle|1\rangle = -|\tilde{a}\rangle|1\rangle$$

$$U_f = (\sum_a |a\rangle\langle a|) \otimes I + (\sum_{\tilde{a}} |\tilde{a}\rangle\langle \tilde{a}|) \otimes X$$

$$\sum_a |a\rangle\langle a| + \sum_{\tilde{a}} |\tilde{a}\rangle\langle \tilde{a}|$$

$$= I \otimes I \otimes \dots \otimes I = I^{\otimes n}$$

Uses: Deutsch-Jozsa, Bernstein-Vazirani, Simon, Grover, counting algorithm

